

Chapter 6 Final gear, differential and axle shaft - Pretest

Time:

11:40

Points:

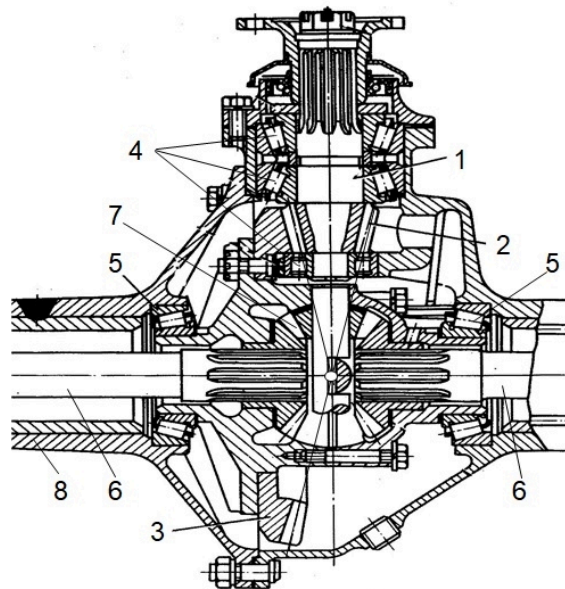
14/26

Correct

1/1 Points

1

Which of the following selections is the final gear in the figure?



Single reduction bevel gear type final gear x

Single reduction spur (cylindrical) gear type final gear

Double reduction center type final gear

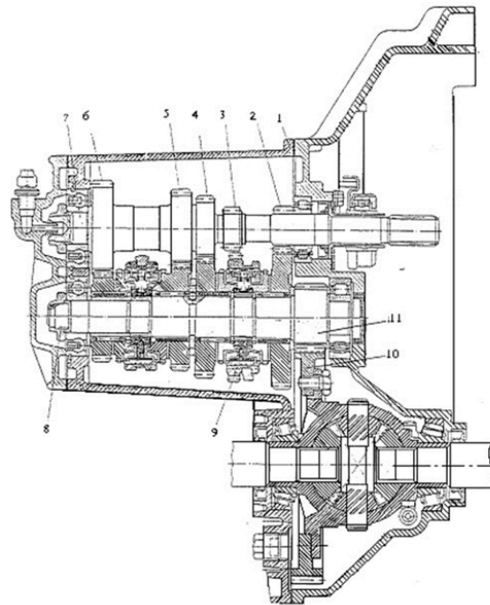
Double reduction side type final gear

Incorrect

0/1 Points

2

Which of the following selections is the final gear in the figure?



Single reduction bevel gear type final gear

Single reduction spur gear type final gear x

Double reduction center type final gear

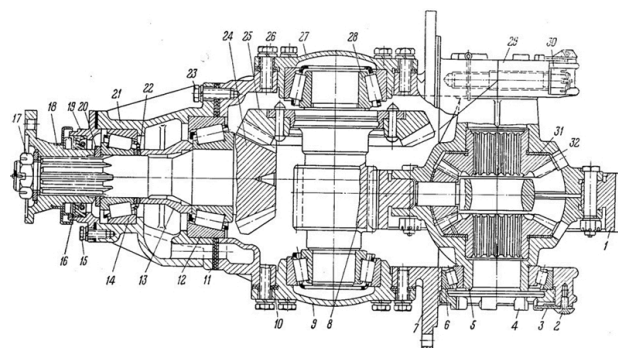
Double reduction side type final gear

Correct

1/1 Points

3

Which of the following selections is the final gear in the figure?



Single reduction bevel gear type final gear

Single reduction spur gear type final gear

Double reduction center type final gear x

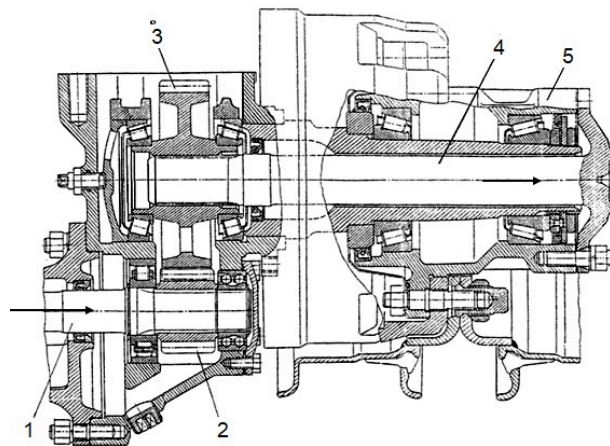
Double reduction side type final gear

Correct

1/1 Points

4

Which of the following selections is the final gear in the figure?



Single reduction bevel gear type final gear

Single reduction spur gear type final gear

Double reduction center type final gear

Double reduction side type final gear x

Correct

1/1 Points

5

Where is the hypoid gear in the following figures?

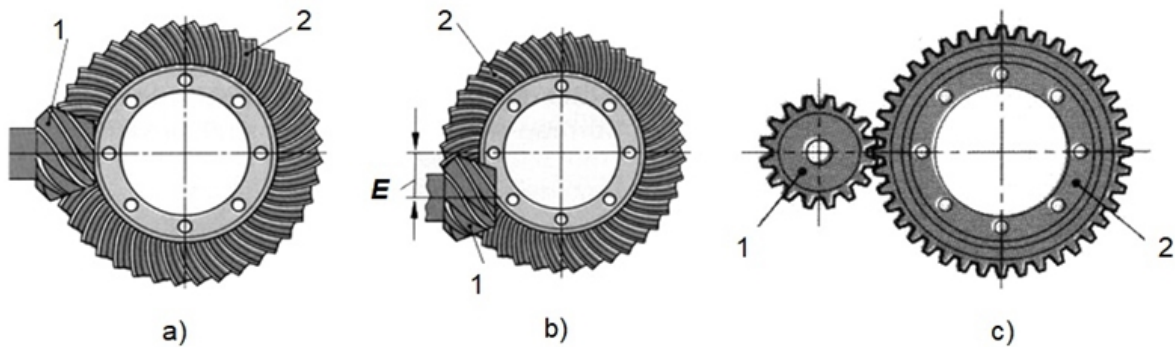


Figure a

Figure b x

Figure c

No figures at all.

Correct

1/1 Points

6

Regarding the advantages of the hypoid gear, which of the following statements is not true?

Has a higher gear ratio and contact ratio than a spiral bevel gear of the same size.

Minimize noise level at high speed.

Can help lower the vehicle's center of gravity while keeping the ground clearance.

Low cost. x

Correct

1/1 Points

7

Regarding the disadvantages of the hypoid gear, which of the following statements is not true?

Complicated manufacturing, requiring high assembly accuracy.

The longitudinal sliding along the tooth length in the meshing area increase.

Requires special lubricating oil

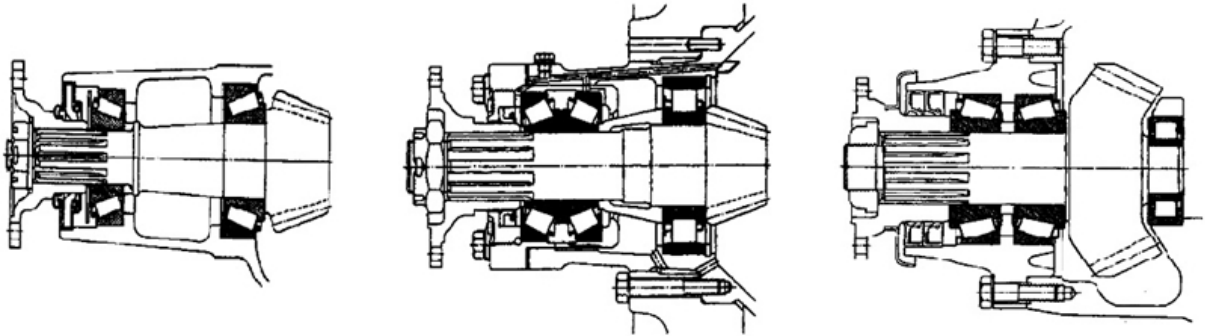
Does not help lower the vehicle's center of gravity. x

Incorrect

0/1 Points

8

Through the layout of bearings, it is possible to evaluate which final gear in the figure has the greatest torque transmission capacity?



Left figure

Middle figure

Right figure x

Both middle and right figure

Incorrect

0/1 Points

9

Regarding the final gear, which of the following statements is not true?

The pinion gear of the FR vehicle is usually arranged in cantilever beam type.

Pairs of two tapered roller bearings are usually arranged in an O letter-shape for better rigidity of the shaft. x

The double reduction final gear is designed for a large gear ratio with little impact on ground clearance.

The double reduction final gear has center type and side type.

Correct

1/1 Points

10

Regarding the functions of differential, which of the following statements is not true?

The differential is used to transmit torque to the drive wheels.

The differential is used to distribute torque to the drive wheels.

The differential is used to change the torque before it is transmitted to the drive wheels. x

The differential ensures that the drive wheels can rotate at different speeds.

Correct

1/1 Points

11

Find the incorrect sentence in the following sentences? Bevel gear type final gears for FR vehicles are usually constructed to:

Be able to adjust the preload of bearing.

Be able to adjust the backlash between pinion gear and ring gear.

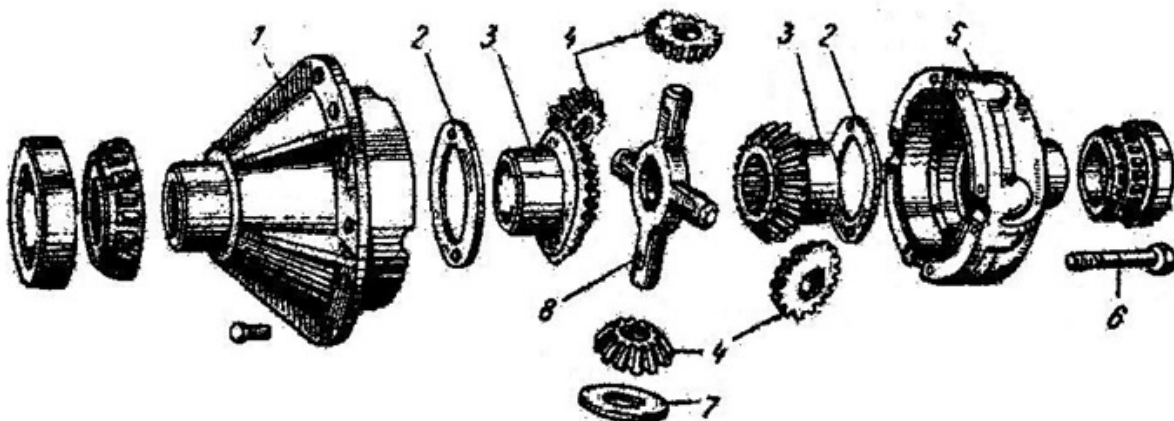
Be able to adjust the vehicle's ground clearance. x

Be able to adjust the tooth contact pattern.

Will be reviewed

12

The figure show the construction of differential, choose the name for numbered part in the following table?



	Pinion shaft	Differential pinion	Side gear	Differential case	Thrust washer
1	☐	☐	☐	☒	☐
2	☐	☐	☐	☐	☒
3	☐	☐	☒	☐	☐
4	☐	☒	☐	☐	☐
8	☒	☐	☐	☐	☐

Correct

1/1 Points

13

In the final gear and differential unit, which of the following part meshinng with the ring gear?

Planetary gear.

Differential case.

Side gear.

Pinion gear. x

Correct

1/1 Points

14

In the final gear and differential unit of FR vehicle, which of the following part meshinng with the pinion gear?

Haft shaft.

Side gear.

Ring gear. x

Differential case.

Correct

1/1 Points

15

Which the following part constructed the final gear and differential of the FR vehicle, is directly related to the speed distribution operation.

Driven gear (ring gear).

Final gear housing.

Side gear. x

Driving gear (pinion gear).

Correct

1/1 Points

16

Talking about full floating rear axle shaft, which of the following sentences is incorrect?

Normally, used for heavy trucks and buses.

Its construction make vehicle can support heavy load.

Rear axle shaft can be removed from axle housing without disassembly wheel.

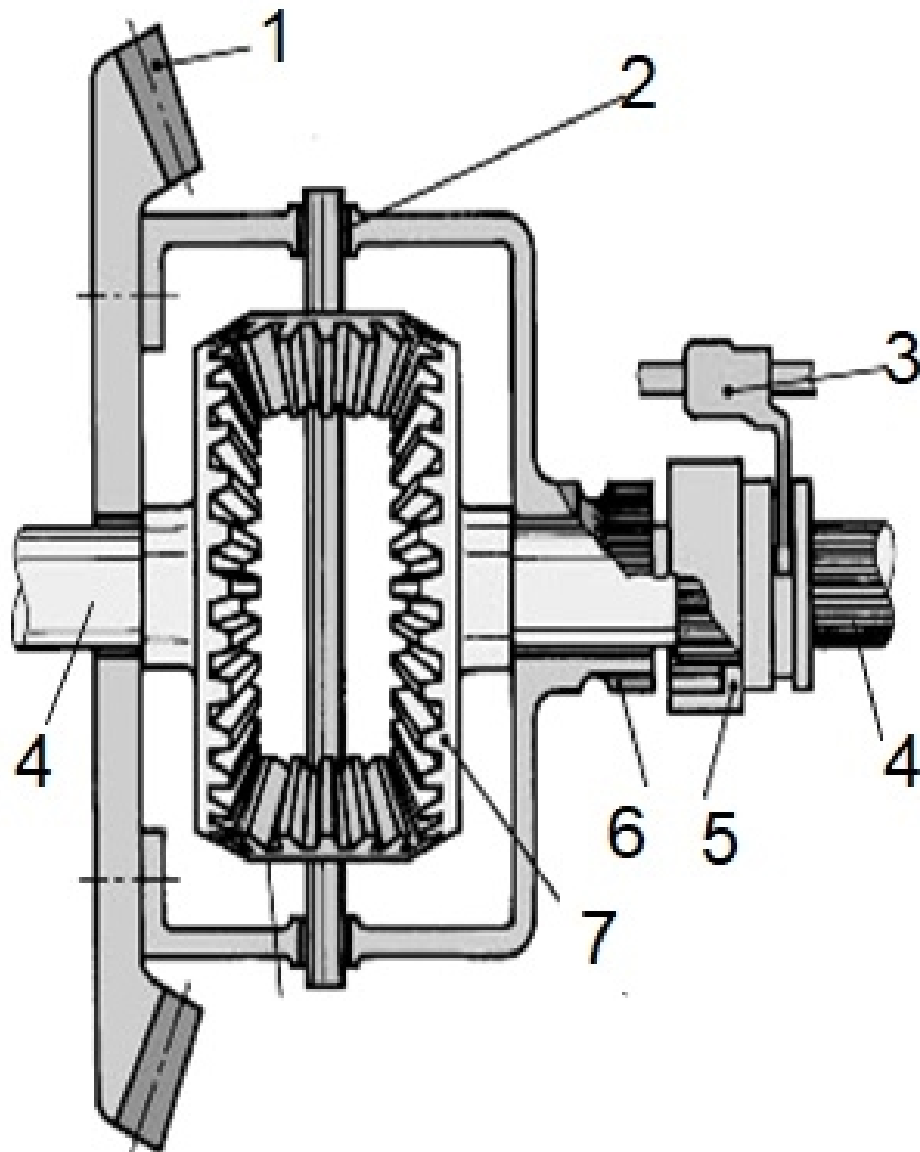
Rear axle shaft both transmits torque to the drive wheels and support the load of vehicle. x

Correct

1/1 Points

17

The following figure shows differential construction with locking differential, find the incorrect statement in the following sentences?



When the one side drive wheel is spined, the differential must be locking differential.

Locking operation is possible when the wheels have come to a complete stop.

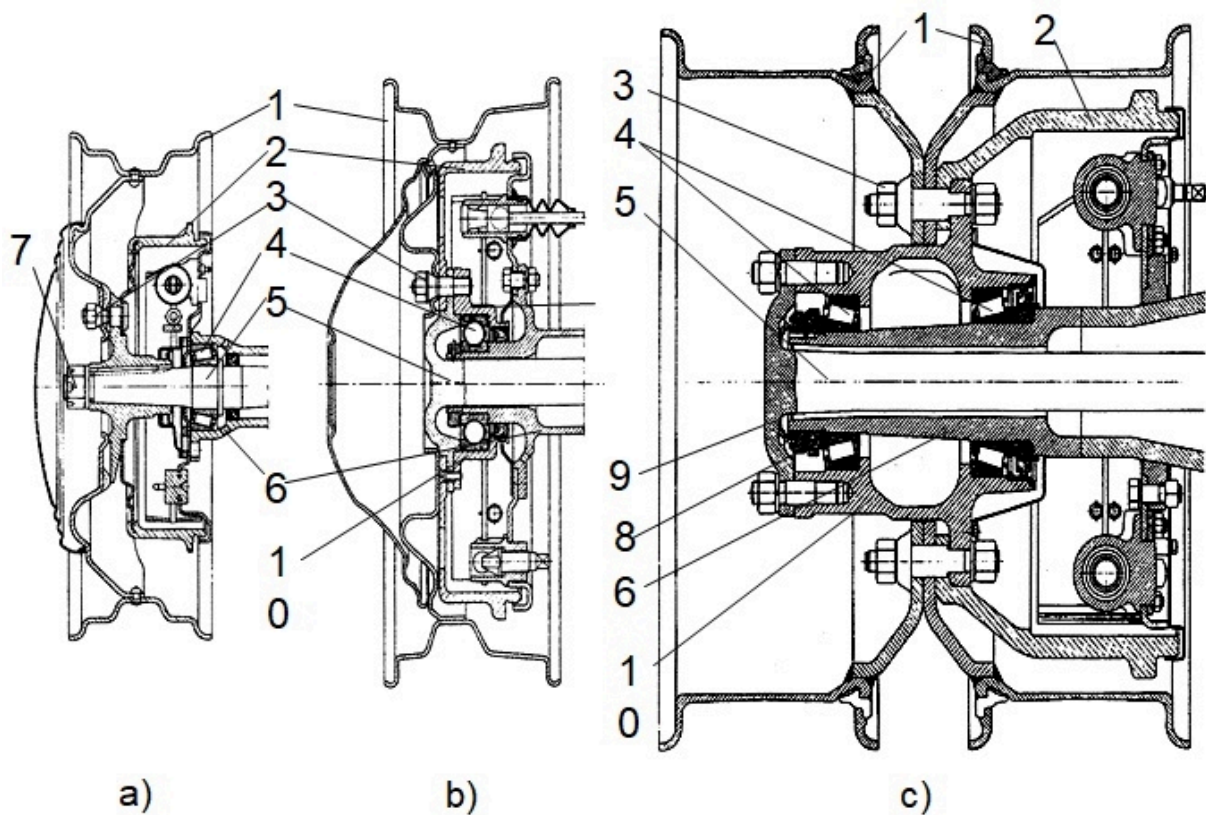
When locking, the driver act the differential lever, the lock shift lerver 3 push the sliding sleeve 5 into the teething ring 6 to lock the differential.

There is usually no locking differential indicator on the dashboard, so remember to unlock it after use. x

Will be reviewed

18

Choose a proper name for the axle shaft in figure a, b, c?



	No floating axle shaft	Semi floating axle shaft (1/2 load)	Semi floating axle shaft (3/4 load)	Full floating axle shaft	Load increasing axle shaft
Figure a	☐	☒	☐	☐	☐
Figure b	☐	☐	☒	☐	☐
Figure c	☐	☐	☐	☒	☐

Incorrect

0/1 Points

19

Talking about semi floating axle shaft, which of the following sentences is correct?

Normally, used for heavy trucks and buses.

In terms of construction, it is possible to make the vehicle withstand a bigger load than the fully floating axle shaft.

There is only one supporting bearing between the axle shaft and the axle housing.

Axle shaft transmits torque to the drive wheel and support only a part of the vehicle's load. x

Correct

1/1 Points

20

Talking about full floating axle shaft, which of the following sentences is incorrect?

Normally, used for heavy trucks and buses.

In terms of construction, it is possible to make the vehicle withstand a bigger load than other axle shaft.

The axle shaft can be pulled out without jacking up the axle or removing the wheels.

The end of axle shaft mounted with the hub has one supporting bearing. x

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